1. **On your Windows Server VM, create a shared folder named 'MusicLibrary' and set the share permissions so only the 'Students' group has read access.** ○

1. **Group Creation (Prerequisite):**
    a. Open **Run** (`Win + R`), type **`lusrmgr.msc`**, and press **Enter** to open *Local Users and Groups*.
    b. Right-click on **Groups**, select **New Group**, set the Group Name as **`Students`**, and click **Create**.
2. **Folder Creation & Share Permissions Setup:**
    a. Create a new folder named **`MusicLibrary`** in the `C:` drive.
    b. Right-click on the folder, select **Properties**, and switch to the **Sharing** tab.
    c. Click on **Advanced Sharing**, check **Share this folder**, and then click on **Permissions**.
3. **Restricting Access to 'Students' Group:**
    a. Select **Everyone** from the list and click **Remove**.
    b. Click **Add**, type **`Students`**, click **Check Names**, and press **OK**.
    c. Under permissions for **Students**, check **Allow** for **Read** access only.
    d. Click **Apply** and **OK** on all open dialog boxes to save the settings.

2. **Configure NTFS permissions on the 'MusicLibrary' folder so that members of the 'Students' group can read and write files, but cannot delete any files.

Hint: Use the Advanced Security Settings to deny delete permissions while allowing modify.**

○

Step 1: Open Advanced Security Settings

1. On your Windows Server, right-click the MusicLibrary folder and select Properties.
2. Go to the Security tab.
3. Click the Advanced button at the bottom to open the Advanced Security Settings window.

Step 2: Add the 'Students' Group and Grant Modify Permission 1. In the Advanced Security window, click the Add button at the top.
2. Click Select a principal, type Students in the box, click Check Names, and then click OK.
3. Set the Type dropdown to Allow.
4. Set the Applies to dropdown to This folder, subfiles and folders.
5. Under Basic permissions, check the box for Modify (this automatically checks Read, Write, and other necessary file-editing permissions).
6. *Do not click OK yet.*

Step 3: Explicitly Deny Delete Permissions

1. In the same permission entry window, click on Show advanced permissions near the bottom.
2. Scroll down the permissions list and locate:
   - Delete
   - Delete subfolders and files
3. Uncheck them under the "Allow" column if checked, or switch the entry to a Deny type (or add a separate explicit Deny entry for Students where Delete and Delete subfolders and files are checked under Deny).
4. Click OK to save the entry.
5. Click Apply and then OK on all remaining open windows.

- **Procedure:**
  1. Opened folder *Properties -> Security -> Advanced*.
  2. Added the *Students* group and assigned Modify permissions (Allow).
  3. Navigated to advanced permissions and explicitly set Delete and Delete subfolders and files to Deny.

3. From a Windows client system, map the 'MusicLibrary' shared folder as a network drive and try to copy, edit, and delete a file. Document which actions succeed or fail, and explain why based on the permissions you set.

1. Mapping the Network Drive (Client System):

- Open File Explorer on the Windows client machine.
- Right-click on This PC and select Map network drive...
- Choose an available drive letter (e.g., Z:).
- Enter the server folder path: \\<Server_IP_Address>\MusicLibrary (e.g., \\192.168.1.50\MusicLibrary).
- Enter the credentials of a user belonging to the Students group when prompted, and click Finish.

2. Testing Operations & Results:

Operation / Action	Result	Status
Copy / Create File (Saving a new file into the share)	Success	Allowed
Edit File (Modifying and saving changes to an existing file)	Success	Allowed

Delete File (Attempting to delete a file or folder)	Failed (Access Denied)	Blocked

3. Explanation (Based on Permissions Set):

- Why Copy & Edit Succeeded: The Students group was explicitly assigned Modify permissions in the NTFS security settings. The *Modify* permission grants users the rights to Read, Write, Create, and Change files and folders, allowing seamless file creation and editing from the mapped network drive.
- Why Delete Failed: In the Advanced Security Settings, Delete and Delete subfolders and files permissions were explicitly configured as Deny for the *Students* group. In Windows NTFS architecture, an Explicit Deny overrides all other permissions. Therefore, even though users have full modification rights, any attempt to delete files or folders is strictly blocked with an *Access Denied* error.

4. Create a second shared folder called 'PhotoShare' and set up share and NTFS permissions so that only your user account has full control. Attempt to access this share from another user account and note the result.

- **Procedure:**

- **Folder Creation:**
  - **Created a new folder named PhotoShare on the Windows Server (e.g., in the C: drive).**
- **Share Permissions Configuration:**
  - **Right-clicked PhotoShare -> Properties -> Sharing tab -> Advanced Sharing.**
  - **Checked Share this folder, clicked Permissions, removed the *Everyone* group, and added the Administrator/owner user account with Full Control (Allow).**
- **NTFS Permissions Configuration:**
  - **Switched to the Security tab -> Clicked Advanced.**
  - **Clicked Disable inheritance and selected *Convert inherited permissions into explicit permissions*.**
  - **Removed all unauthorized users and groups (such as *Users*, *Everyone*, *Students*), leaving only the administrator account with Full Control.**

- **Clicked Apply and OK to save.**

2. Testing & Result:

- **Action: Attempted to access the share from a standard or different user account using the network path \\<Server_IP>\PhotoShare.**
- **Result: Windows immediately blocked the connection and displayed an "Access Denied" error message.**
- **Conclusion: The strict restriction successfully prevents any unauthorized user from viewing or accessing the contents of the PhotoShare folder, verifying that the share and NTFS security isolation is working as intended.**